Lab 3 - Log Transformation

June 7, 2025

1 Theory

The log transformation is a point processing technique used in image processing to enhance the details in the darker regions of an image while compressing the brighter regions. The transformation is defined as:

$$s = c \cdot \log(1 + r)$$

where: - (r) is the input pixel value (normalized between 0 and 1), - (s) is the output pixel value, - (c) is a scaling constant.

This transformation is particularly useful for images with a large dynamic range, as it makes low-intensity values more distinguishable.

[8]:
```
import numpy as np
from PIL import Image
import matplotlib.pyplot as plt
```

[9]:
```
def log_transform(input_image):
  # resizing image
  img = input_image.resize((400,400), Image.Resampling.LANCZOS)
  # convert to numpy array
  numpy_image = np.array(img)
  numpy_image = numpy_image/255
  numpy_image = numpy_image + 1
  numpy_image = np.log(numpy_image)
  print(type(numpy_image))
  numpy_image = numpy_image * 255
  numpy_image = np.around(numpy_image,decimals=0)
  log_image = Image.fromarray(numpy_image)
  log_image = log_image.convert("L")


    #plotting input and output images
  # set up side-by-side image display
  fig = plt.figure()
  fig.set_figheight(15)
  fig.set_figwidth(15)
```

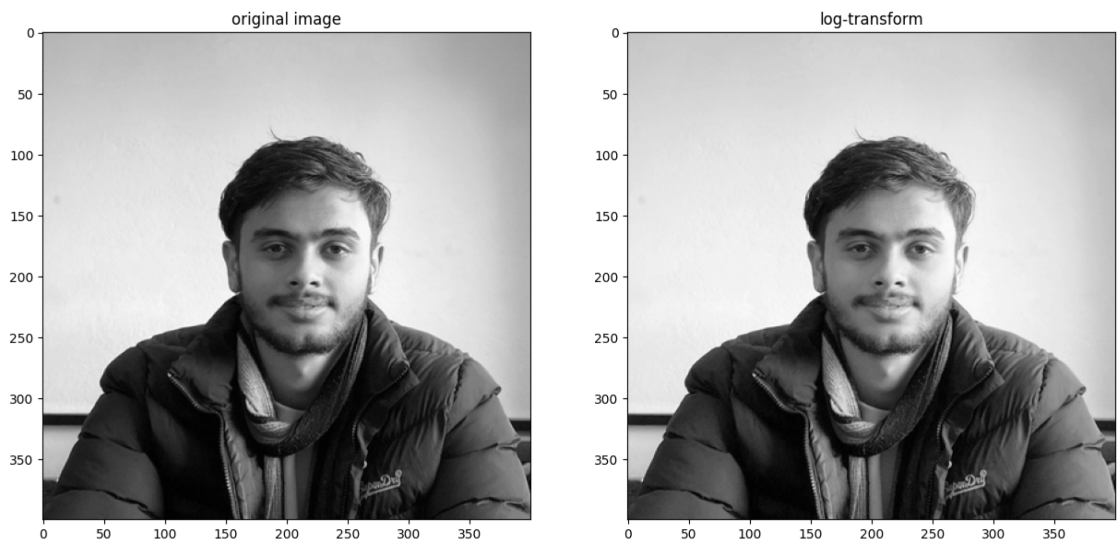
```python
    fig.add_subplot(1,2,1)
    plt.imshow(img, cmap='gray')
    plt.title('original image')

    fig.add_subplot(1,2,2)
    plt.imshow(log_image, cmap='gray')
    plt.title('log-transform')

    return log_image
```

[10]:
```python
# reading image and converting to gray scale
img = Image.open('img/samir.jpg').convert('L')
# display image
a = log_transform(img)
```

```
<class 'numpy.ndarray'>
```



2 Conclusion

The log transformation effectively enhances the visibility of details in the darker areas of an image while compressing the brighter regions. This technique is valuable for preprocessing images with high dynamic range, enabling better feature extraction and analysis in subsequent image processing tasks.